

Production Source-Claim Inventory: A Closed-World Census for Local Occurrence Edges

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Abstract

The first missing object in MVP7 is a source-backed local actual-occurrence edge family. The earlier TPC-163 census froze only eleven artifacts from six papers and explicitly was not a future-schema scanner. We now freeze a larger, contiguous theorem corpus: exactly one `main.tex` for every TPC paper from TPC-133 through TPC-172. Machine-readable production archives and route outputs are locked as a separate data/status substrate and are never promoted to theorem evidence.

Every theorem file is classified as mapped and disqualified, reviewed with no candidate statement, not mapped yet, or qualifying. The exact partition is 30, 10, 0, 0. Thirty-two occurrence-adjacent claims are mapped with source path, canonical hash, theorem and formula locators where present, derivation AST, carrier and exact qualification gaps. None concludes a local actual-occurrence edge above the five-field production cut address with exact edge weight, fixed $h_0 = 2$, and physical-normalization lineage. Thus the qualifying inventory is empty in this frozen declared corpus. This is a closed-world L1 source-census obstruction, not mathematical nonexistence and not arithmetic L2 progress.

1 Declared corpus

Definition 1.1 (Frozen theorem corpus). *Let $\mathcal{C}_{173}$ contain the canonically normalized `main.tex` file in the unique TPC- n directory for every*

$$133 \leq n \leq 172.$$

Thus $|\mathcal{C}_{173}| = 40$. The selection is contiguous and number-based; it is not a keyword sample.

The interval begins at the executable native entrance and contains the cut compiler, occurrence contracts, formal obstructions, return theorems, source censuses, local-gluing theorem, phase-metric corridor and MVP7 decision [1–3]. Twenty-two production archives and route outputs are additionally hash-locked as *data/status substrate*. A substrate status can say that a witness is absent; its digest cannot prove a mathematical theorem.

2 Qualification contract

Definition 2.1 (Qualifying local edge claim). *A claim qualifies only if it has all of:*

1. *a canonical source path and recomputed source hash;*
2. *resolving theorem and formula locators;*
3. *a nonempty derivation AST;*
4. *a conclusion asserting an actual local occurrence edge;*
5. *the source address $(\ell, k, d_{\text{nat}}, j_L, j_K)$;*

6. an exact edge weight; and

7. fixed $h_0 = 2$ and physical-normalization lineage.

Actual active support and canonical/minimal actual representation are not silently added to this contract. TPC-166 proves that they are independent H1 roots. Closing a local-edge root would not close either of them.

3 Exact corpus partition

Theorem 3.1 (Complete inventory). *The forty files in $\mathcal{C}_{173}$ have the exact partition*

<i>Disposition</i>	<i>Files</i>
<i>MAPPED_DISQUALIFIED</i>	30
<i>REVIEWED_NO_CANDIDATE</i>	10
<i>NOT_MAPPED_YET</i>	0
<i>QUALIFYING</i>	0

(1)

and the thirty-two mapped occurrence-adjacent claims contain no claim satisfying [definition 2.1](#). Consequently

$$\boxed{\# \mathcal{Q}_{173} = 0.} \quad (2)$$

Proof. The generator resolves exactly one TPC directory and one `main.tex` for each integer from 133 through 172, recomputes all canonical hashes, and demands one disposition per file. Every mapped claim resolves its nonnull locators and is evaluated against the nine qualification fields. The ten remaining papers are explicitly reviewed and have no statement purporting to create a production local occurrence edge. No file remains in NOT_MAPPED_YET. Every mapped claim has at least one recorded semantic or data gap, so the qualifying list is empty. $\square$

4 Why the near-claims do not qualify

The strongest near-claims are source-located, but their carriers differ:

Claim family	Carrier	Exact gap
TPC-153 shadow identities	partial cut-occurrence shadow	the row is explicitly not an actual occurrence
TPC-154 obstructions	formal completion fibre	formal children and current-artifacts-only scope
TPC-155 soundness	supplied witness	no production witness; external theorem truth is not verified
TPC-164 minimum key	frozen cut archive	archive address, not occurrence identity or edge
TPC-165 gluing	supplied compatible formal local families	local families are hypotheses and production instantiation is absent
TPC-149, 167–170	actual determinant-two arithmetic core	no five-field cut-address-to-occurrence edge or edge lineage

In particular, “actual core” on the arithmetic branch does not supply a structural edge from a frozen cut row. Conversely, an exact shadow edge of weight one does not acquire actual-carrier semantics.

5 Scoped maximal conclusion

Proposition 5.1 (Closed-world boundary). *Equation (2) means only that no qualifying declaration occurs in $\mathcal{C}_{173}$. It does not imply that no local actual-occurrence edge exists, that an enlarged corpus is empty, or that the selected architecture is infeasible.*

Proof. The corpus is finite and explicitly declared. No completeness theorem for all mathematical sources is assumed. New source mathematics or an explicitly enlarged corpus can change the inventory. $\square$

The next object is therefore a minimal source-locked local-edge witness contract. That contract can be proved nonvacuous synthetically, but it cannot turn the empty production inventory into an edge family.

6 Claim firewall

This paper is an L1 source-census obstruction with finite L0 machinery. It proves no production local occurrence family, actual active support, canonical/minimal actual representation, named fixed phase, positive fixed- X L2, strict $1/400$, prime-pair lower bound, or twin-prime theorem.

References

- [1] Liang Wang. Source-locator census for the h1 crosswalk. *TPC Research Archive*, 2026. TPC-163.
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- [3] Liang Wang. Mvp7 on the occurrence–phase–atomic corridor. *TPC Research Archive*, 2026. TPC-172.